

Evaluating the Value of Topology-Aware Features for Cyber-Physical Grid Monitoring: Detection, Localization, and End-to-End Latency

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Abstract—Distinguishing cyber false-data-injection attacks (FDIAs) from legitimate physical disturbances requires accurate detection, localization, and sufficiently fast end-to-end processing. We evaluate the marginal value of explicit topology-relational features across a synthetic 5-bus microgrid and standard IEEE 14-bus and 30-bus systems using deployable WLS residuals, validation-only model selection, and a matched topology-free residual-plus-prior ablation. Across 16 independent seeds, topology features are neutral on the 5-bus system on both tasks but produce small positive gaps (roughly 0.01–0.03 balanced-accuracy/top-1-accuracy points) on both detection and localization at both IEEE systems; three of the four IEEE gaps remain significant under a Bonferroni correction, one does not. Separately, pipeline profiling reduces median end-to-end latency from 42.98 ms to 16.2–17.0 ms, with p99 at 17.0–19.2 ms but rare maxima of 89.1–92.7 ms. Finally, neither feature set generalizes to a completely unseen attack type (0–1.3% recall versus 91–100% when represented in training).

Index Terms—false data injection attack, state estimation, cyber-physical systems, attack localization, latency profiling, feature ablation, inverter-based resources

I. INTRODUCTION

Modernizing power grids toward higher penetrations of inverter-based resources (PV, battery storage, grid-forming converters) changes the character of protection and monitoring in two ways relevant here. First, inverters provide far less fault current than synchronous generators, which strains protection schemes designed around synchronous-machine fault signatures [1]. Second, the same measurement infrastructure used for state awareness is now also an attack surface: an adversary who can spoof a subset of sensor readings can attempt a false-data-injection attack (FDIA) against the state estimator that is, by construction, statistically indistinguishable from a legitimate measurement under a residual-based bad-data test [2].

Recent incidents illustrate the operational relevance of cyber monitoring in renewable-heavy grids: a December 2025 cyber-attack, attributed to the Dragonfly/Static Tundra threat actor, damaged controller firmware and deployed wiper malware across more than 30 Polish wind and solar farms and a combined heat-and-power plant serving nearly half a million customers, though production and supply were ultimately not

disrupted [3]. That incident was an OT/firmware-integrity attack rather than a state-estimation FDIA, but it illustrates the need to distinguish malicious anomalies from physical events under operational time constraints.

Operational monitoring therefore combines three questions: whether an anomaly is cyber or physical, where it originates, and whether a decision can be produced within a relevant computational budget. These questions are often treated separately. FDIA detection is commonly evaluated offline on fixed datasets; attack localization is studied less often; and end-to-end computational latency has received comparatively limited attention. A recent benchmark [4] found that deep classifiers for joint cyber-physical discrimination can be individually fast (sub-15 ms) while the measured end-to-end pipeline remains much slower (50–90 ms). In parallel, topology-aware methods ask whether explicit relational information between electrically connected measurements improves detection or localization beyond topology-free features.

We evaluate detection and localization across three networks and separately profile end-to-end computational latency on the 5-bus implementation. The evaluation enforces separation between relational and non-relational feature sets, uses deployable WLS residuals, and selects models exclusively on validation data. The latency study reproduces the qualitative finding of a fast classifier embedded in a slower pipeline [4] and identifies a specific software bottleneck: rebuilding a fixed-structure measurement table every cycle. Replacing repeated object creation with in-place value updates accounts for most of the measured latency reduction.

A. Contributions

- A 16-seed cross-network ablation showing small positive detection and localization gaps for explicit topology-relational features at both standard IEEE test systems (14-bus, 30-bus), but no measurable benefit on either task at the small custom 5-bus microgrid (Sections IV and VII), together with a quantitative redundancy analysis of the 5-bus result.
- End-to-end latency profiling that identifies measurement-table reconstruction as a major software bottleneck and

reduces median latency from 42.98 ms to 16.2–17.0 ms; p99 remains below the 20 ms reference in both repeated runs, while rare maxima do not (Section V).

- A zero-day stress test showing that recall collapses when an entire attack family is withheld from training (Section VI).

II. RELATED WORK

A. Foundational FDIA and Stealth Constructions

The threat model follows Liu, Ning & Reiter’s foundational FDIA construction [2], adapted here to the AC case: an attacker with network-model knowledge constructs a measurement perturbation that preserves the WLS residual. Later work considers progressively weaker attacker knowledge: stealthy attacks constructible without topology or line-reactance information [5], unknown line parameters specifically [6], and topology-blind generative or physics-informed attack synthesis [7]. Low-magnitude attacks near the sensor-noise floor are also studied explicitly [8]; the hardened 5-bus setting in Section IV targets this more difficult regime.

B. Deep-Learning FDIA Detection

CNN, LSTM, and transfer-learning architectures have been applied extensively to FDIA detection [9]–[11], typically as binary classification on a fixed attack distribution; a separate line of work instead uses physics-informed neural networks for secure state estimation without adversarial training [12]. The `residual_only` and `prior_only` baselines used here are consistent with this setting. The present ablation isolates the marginal contribution of explicit topology-relational features beyond their topology-free combination, under deployable feature construction and validation-only model selection.

C. Topology-Aware and Graph-Based State Estimation

Topology-aware graph neural network (GNN) methods use graph convolution or attention to exploit network structure in state estimation [13]–[16]. Prior work presents a mixed picture. [13] and an earlier, closely related study by an overlapping author group [14] compare their GNN against a same-feature “regular” DNN baseline (identical PMU-derived node features, no adjacency matrix), providing a correctly scoped topology ablation; [15]’s non-graph baseline, by contrast, does not clearly document an input-matched comparison. These studies show that topology ablation is not absent from the literature. We instead ask a complementary question: whether explicitly relational features add information beyond a matched topology-free feature set before model architecture becomes a confounding factor. Section IV also evaluates the linear recoverability of relational features from the topology-free representation as a model-independent redundancy diagnostic.

D. Attack and Event Localization

Localization work spans capsule networks for load-altering (not FDIA) attacks evaluated across 14/39/57-bus systems [17], moving-target-defense-assisted localization [18], and on-line detection-plus-localization schemes [19]. The cited localization studies do not explicitly document sensitivity to

network topology/meshedness or an ablation of relational features against a matched non-relational baseline, which is the comparison examined in Sections IV and VII.

E. Latency-Aware Detection and Classification

[4] provides the closest latency comparison: on a larger industrial electromagnetic-transient (EMT) testbed, classifier inference is fast while end-to-end latency remains substantially higher. Section V examines the same pipeline-level question and isolates a software-level bottleneck in the present implementation.

F. Generalization to Unseen Attack Types

Generalization to attack families absent from training remains an important deployment question. Section VI evaluates this directly for the two FDIA families used here.

G. Broader Gaps Identified in Reviews

A 2025 scoping review of machine-learning applications in power-system protection [20] identifies two relevant gaps: limited reporting of computational latency and little cross-topology evaluation. Sections V and VII address these issues within the scope of the present test systems.

III. TESTBED AND METHODOLOGY

A. Network

A 5-bus, 0.4 kV synthetic microgrid built in `pandapower`: point of common coupling, a PV bus modeled as a steady-state grid-following (GFL) power injection, a BESS bus modeled as a steady-state grid-forming (GFM) power injection, and two load buses, connected by 4 lines. Both are static `sgen` operating-point injections, not dynamic converter-control models – no droop control or virtual inertia is simulated; the GFL/GFM labels describe each device’s intended role in the microgrid, not modeled control dynamics. `net.f_hz = 50.0` (verified from the model object, not assumed – this fixes the one-cycle protection-relevant computational target below to 20 ms).

B. State Estimation

State estimation uses weighted least squares (WLS) via `pandapower.estimation.estimate`, with 23 measurements per cycle: voltage magnitude and P/Q injection at all 5 buses, plus P/Q from-side flow on all 4 lines. This is a conventional model-driven state-estimation setting of the type reviewed in [21]. Across all three test systems, residual features are the deployable normalized measurement residual $(z - h(\hat{x}))/\sigma$ computed from the WLS solution; no feature is constructed from an unobservable ground-truth state.

C. Threat Model

Two synthetic attack families, both targeting a randomly chosen load bus per scenario:

- *Naive*: a single voltage-magnitude sensor is spoofed by a fixed-range random offset.
- *Model-consistent (“stealth”) FDIA*: following the Liu–Ning–Reiter construction adapted to the AC case, the true

state is perturbed at the target bus and the corresponding measurement-space attack vector $a = h(x_{\text{attack}}) - h(x_{\text{true}})$ is computed exactly, so the attack is, by construction, invisible to the chi-square residual test.

A benign hard-negative physical load disturbance (a 25–70% load step at a randomly chosen bus) is included specifically to be confusable with an attack in raw residual terms.

D. Prior Construction

Every replication starts from one randomly sampled base operating point (a load/PV level, shared unmodified across that replication’s 4 scenarios). Before any attack or physical disturbance is applied to build the “true” network for a given scenario, an independent copy of that same base operating point has its load values perturbed by a fresh, independently drawn forecast error ($\mathcal{N}(0, 2.5\%)$ per load at 5-bus, clipped to $\pm 6\%$; $\mathcal{N}(0, 5\%)$ at IEEE 14-bus/30-bus, a cross-network noise-model difference documented as a limitation below). This perturbed copy is then solved through its own, separate AC power flow – the result is the “prior.” `prior_only` features are the deviation between the WLS state estimate (computed from the scenario’s actual, possibly-attacked measurements) and this prior’s own voltage/angle solution. The prior is constructed *before*, and entirely independently of, whichever scenario (clean, naive attack, stealth attack, physical disturbance) that replication’s true network becomes – it never sees the true or attacked state, only an independently-perturbed forecast of the same nominal base point. As a result, stealth attacks, invisible in residual space by construction, become visible: the attack is only tuned to be consistent with the WLS residual test, not with an estimate the defender never let it see.

E. Feature Sets

Four feature sets are compared. `residual_only`: conventional WLS measurement residuals. `prior_only`: the prior-deviation features defined above. `residual_plus_prior`: the two combined, with no relational or structural information at all – the topology-free ablation. `topology_fusion`: `residual_plus_prior` plus explicit relational features describing each bus relative to its network neighbors (local-minus-neighbor innovation/residual, incident-edge aggregates); no static structural metadata (node degree, device role) beyond that, since neither is a scenario-specific neighbor comparison and both risk leaking attack-target eligibility (the attack generator above only ever targets load buses). To preserve the intended ablation, non-relational baselines exclude all relational columns, and static node-degree/device-role metadata is excluded from every feature set because it could encode attack-target eligibility when attacks are restricted to load buses.

F. Models and Evaluation

LogisticRegression, RandomForest, and HistGradientBoosting classifiers are compared with replication-group-safe 70/15/15 train/validation/test splits and group-aware bootstrap confidence intervals. Model selection uses the validation split

TABLE I
TOPOLOGY_FUSION VS. THE CORRECTED, TOPOLOGY-FREE ABLATION
(BEST VALIDATION-SELECTED MODEL PER CELL; 5-BUS, $n=300$
DETECTION / $n=150$ CYBER-ONLY LOCALIZATION TEST SCENARIOS)

Condition	Detection (bal. acc.)		Localization (top-1)	
	topo. fusion	res.+prior	topo. fusion	res.+prior
Hard	0.950	0.937	0.967	0.967
Harder ($+2\times$ noise)	0.950	0.947	0.960	0.967

exclusively: each candidate model is scored on validation, the best-scoring model per feature set is selected, and only that model’s (until then unseen) test-set score is reported.

G. Sample-Size Terminology

A *replication* is one independently sampled base operating point (load level and forecast draw). Each replication produces four matched *scenarios*: clean, naive attack, stealth attack, and physical disturbance. Thus, N replications yield $4N$ scenarios. The 70/15/15 split is applied at the replication level so that matched scenarios from the same operating point cannot cross train, validation, and test partitions. With $N=500$, this produces approximately 300 test scenarios for detection and 150 cyber-only test scenarios for localization. Unless otherwise specified, n denotes the resulting test-scenario count rather than the number of replications.

IV. 5-BUS ABLATION AND REDUNDANCY ANALYSIS

The primary 5-bus evaluation uses a harder attack regime with perturbation magnitude reduced by roughly $3\times$ toward the sensor-noise floor and is evaluated on $n=300$ test scenarios (Section III). `prior_only` alone is missing residual information that `topology_fusion` has access to, so a comparison against `prior_only` specifically cannot, by construction, distinguish “topology helped” from “having residual information at all helped”; the fair comparison is against the topology-free `residual_plus_prior` ablation instead.

A. Primary Ablation

Table I compares `topology_fusion` against `residual_plus_prior` (Section III), for the primary seed and both stress conditions.

Detection shows a small edge for `topology_fusion` at this one seed, in both conditions ($+0.013$ Hard, $+0.003$ Harder); localization is an exact tie under Hard and a small edge for `residual_plus_prior` under Harder (-0.007). A single seed is insufficient for inference; the 16-seed replication below provides the primary comparison. Both feature sets also outperform `prior_only` alone (0.860–0.870 detection, 0.687–0.693 localization) and `residual_only` alone (0.700 detection, 0.547 localization), indicating that the dominant improvement comes from combining residual and prior information rather than from topology.

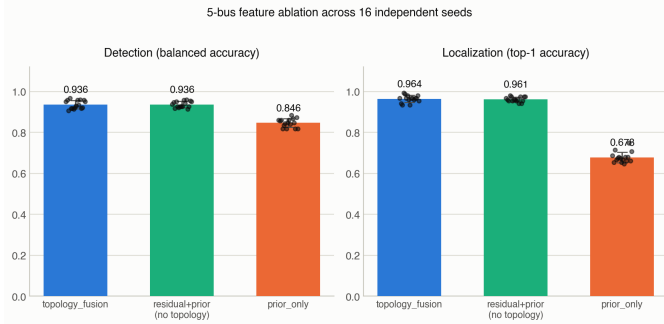


Fig. 1. Topology fusion vs. the topology-free residual+prior ablation and the prior-only baseline across 16 independent random seeds (5-bus, hard attack magnitude). Bars show mean $\pm$ standard deviation; dots show individual seed results. $n=300$ detection / $n=150$ cyber-only localization test scenarios per seed.

B. Multi-Seed Replication

Figure 1 summarizes the 16-seed result. With localization training restricted to cyber-scenario nodes only (matching the convention used at IEEE 14-bus/30-bus, Section VII), the `topology_fusion`-vs-`residual_plus_prior` detection gap is -0.000 ± 0.009 (range -0.013 to $+0.017$, 95% CI $[-0.005, +0.004]$), and the localization gap is $+0.003 \pm 0.017$ (range -0.020 to $+0.027$, 95% CI $[-0.006, +0.012]$). Both confidence intervals cross zero, consistent with a near-zero effect on this testbed for both tasks.

C. Redundancy Diagnostic

The near-zero 5-bus gaps are consistent with high linear recoverability of the relational features from the local feature set. Predicting each of the 65 relational columns from `residual_plus_prior` alone, with 5-fold cross-validated ridge regression, gives a mean out-of-fold R^2 of 0.956 (median 1.000, minimum 0.613, 90.8% of columns above 0.8): a neighbor-mean or incident-edge aggregate over a fixed, small neighbor set is, on a network this size, close to an affine function of the same per-bus values `residual_plus_prior` already provides in full. A sufficiently flexible classifier (Hist-GradientBoosting, the best model throughout) may therefore reconstruct much of the relational information internally, consistent with the near-zero 5-bus gaps. This explanation is specific to the fixed 5-bus topology and does not predict the larger-network results in Section VII, where a separate size-agnostic feature design is used. The Discussion evaluates whether reduced redundancy can explain the observed cross-network pattern.

V. END-TO-END LATENCY PROFILING

Following [4], we measure the full per-cycle decision pipeline (state estimation $\rightarrow$ feature computation $\rightarrow$ classification) against a one-cycle reference of 20 ms at the network’s confirmed 50 Hz. The 20 ms value is a computational reference rather than a universal protection deadline. The experiment evaluates empirical deadline performance, not hard real-time schedulability or an RTOS guarantee. One-time network and model setup is excluded.

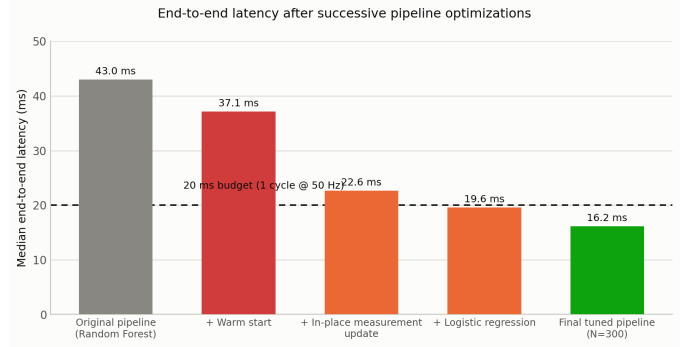


Fig. 2. End-to-end latency after successive pipeline optimizations. In-place measurement updates account for most of the reduction; the lighter classifier provides a smaller additional improvement.

A. Initial Measurement

Median 42.98 ms, $2.15\times$ the 20 ms reference, using the original pipeline (RandomForest classifier, `tolerance=1e-7`, `maximum_iterations=40`, flat state-estimator initialization every cycle).

B. Warm-Start Test

Warm-starting the state estimator from the previous cycle’s converged solution produced only a $1.16\times$ speedup (37.14 ms), ruling out “far from the solution” as the dominant cost.

C. Profiling

Splitting the state-estimation call into its constituent steps showed the measurement-submission step cost 19.9 ms – essentially the same as the estimator’s own solve (19.4 ms). An ablation on `maximum_iterations` ($40 \rightarrow 5$: no speed change; $40 \rightarrow 3$: convergence failure in 19/20 trials) showed the true iteration requirement is roughly 4–5 steps regardless of the 40-iteration cap, ruling out iteration count as the estimator-side cost driver.

D. Measurement-Update Optimization

The measurement table was being rebuilt from scratch every cycle via a general-purpose element-creation API, called once per channel in a Python loop – even though the table’s structure is identical every cycle; only the values change. Replacing this with a one-time table construction followed by an in-place overwrite of the value column each cycle gave a $1141\times$ speedup on this sub-step (19.4 ms $\rightarrow$ 0.017 ms), verified to produce a bit-identical state-estimation result (max difference 0.0).

E. Final Benchmark

With a lighter classifier (LogisticRegression, competitive with RandomForest per Section IV’s accuracy results and faster at inference), two independent $N=300$ reruns on the same documented environment, both with 0 convergence failures, give 16.155–17.015 ms median, 16.429–18.211 ms p95, 16.952–19.190 ms p99, and 89.067–92.657 ms max end-to-end

TABLE II
RECALL ON A WITHHELD VS. TRAINED-ON ATTACK TYPE ($n=75$ TEST
SCENARIOS PER CELL, BOTH CONDITIONS)

Attack type	Features	Withheld	In training
Naive corruption	topo. fusion	0.000	0.907
Naive corruption	res.+prior	0.000	0.907
Stealth FDIA	topo. fusion	0.013	1.000
Stealth FDIA	res.+prior	0.013	1.000

latency (Fig. 2 shows the most recent run). Median, p95, and p99 remain below the 20ms reference in both reruns, while the maximum exceeds it in both. Across the same two reruns, stage medians are 15.855–16.664ms for WLS state estimation, 0.050–0.057ms for post-estimation residual/innovation calculation, 0.083–0.088ms for feature computation, 0.0068–0.0076ms for array assembly, and 0.160–0.200ms for model inference; the WLS solve therefore dominates end-to-end latency. The benchmark environment is Apple arm64 running macOS 15.6.1 with Python 3.12.4, pandapower 2.14.10, NumPy 1.26.4, pandas 2.2.2, and scikit-learn 1.4.2.

The timing benchmark uses a separate 35-feature representation (7 local-plus-neighbor statistics per bus $\times$ 5 buses), rather than `topology_fusion` or `residual_plus_prior` verbatim. Feature computation is a small fraction of total latency, so this difference does not affect the bottleneck conclusion. The classifier is trained on 150 warm-up scenarios generated through the same simulation pipeline (83 attack, 67 clean). The timer includes state estimation, residual/innovation calculation, feature computation, array assembly, and inference.

F. Positioning

[4] reports sub-15ms classifier inference embedded in 50–90ms end-to-end pipelines on an industrial-grade EMT simulator and recommends hardware acceleration. In the present implementation, a large fraction of the latency gap instead comes from avoidable software overhead, indicating that pipeline profiling should precede hardware acceleration. A knowledge-distilled LightGBM detector for virtualized microgrids likewise reports only 54–67ms per 1000 samples (~ 0.06 ms/sample) for inference [22], comparable in scale to the 0.16–0.20ms inference step here and far below the 15.9–16.7ms state-estimation step.

VI. WITHHELD ATTACK-TYPE GENERALIZATION

Zero-day generalization is evaluated by withholding one entire attack family from training and validation, using the same HistGradientBoosting hyperparameters as the 5-bus detector. Recall is then measured on that withheld family under a symmetric test protocol (Table II, Fig. 3). The experiment is repeated for both `topology_fusion` and the topology-free `residual_plus_prior` ablation.

The detector achieves high recall on attack types represented in training but fails to extrapolate to a withheld attack mechanism, and this behavior is identical with and without

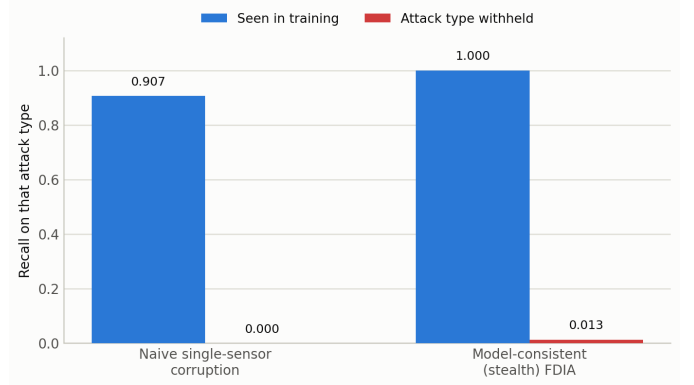


Fig. 3. Recall for attack types represented in training versus withheld from training (topology fusion shown; residual+prior is behaviorally identical, Table II).

topology-relational features: the two feature sets are indistinguishable on every cell of Table II, confirming Section IV’s prediction that they are behaviorally interchangeable, here for a question – generalization to an unseen mechanism – that has nothing to do with topology specifically. Within the two-family attack taxonomy used here, no zero-day generalization is observed for either feature set. This complements, from the opposite direction, a related generalization gap documented in the fault-classification literature: an XGBoost-based classifier trained and evaluated on the *same* distribution still drops from 96.8% (synthetic data) to 85.2% (physical data) [23] – i.e., even without an unseen attack *type*, a synthetic-to-real distribution shift alone costs over 11 points. Both gaps matter for deployment: attack-family shift and synthetic-to-real distribution shift should each be evaluated before fielding a detector. A deployment-aware zero-day evaluation in the adjacent EV/V2X charging-infrastructure domain [24] is a useful methodological reference for what a more rigorous version of Section VI’s test could look like.

VII. CROSS-NETWORK EVALUATION

Sections IV–VI use the custom 5-bus microgrid. Cross-network evaluation is performed on IEEE 14-bus and IEEE 30-bus systems using a separate size-agnostic feature design.

A. Test-System Selection

IEEE 14-bus and IEEE 30-bus were selected because their pandapower bus indexing maps directly to the internal power-flow representation used by the evaluation pipeline (confirmed directly: `net.bus` count matches the internal power-flow-case bus count for both), and they are, independently, among the most commonly used benchmarks in the power-system state-estimation/FDIA literature specifically.

B. Cross-Network Method

All WLS/measurement/attack machinery is network-size-agnostic by construction and was reused unmodified. The original fixed per-bus feature columns do not generalize past 5 buses and were replaced with a size-agnostic design: global

TABLE III

5-BUS VS. IEEE 14-BUS VS. IEEE 30-BUS, ALL 16-SEED MEANS, AT MATCHED TEST-SET SIZES ($n=300$ DETECTION / $n=150$ CYBER-ONLY LOCALIZATION SCENARIOS PER NETWORK)

	topo.	res.+prior	prior	res.
Det., 5-bus	0.936	0.936	0.846	0.680
Det., IEEE-14	0.882	0.873	0.705	0.663
Det., IEEE-30	0.877	0.849	0.680	0.646
Loc., 5-bus	0.964	0.961	0.678	0.568
Loc., IEEE-14	0.925	0.907	0.531	0.462
Loc., IEEE-30	0.901	0.888	0.510	0.430

Per-seed gap statistics and uncertainty are reported in Table IV.

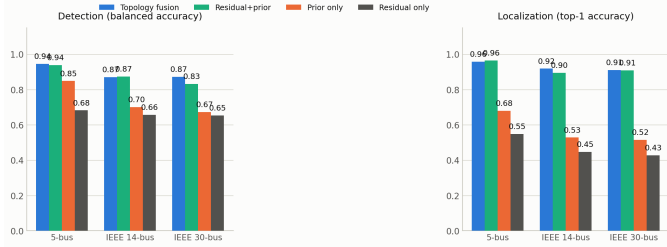


Fig. 4. 5-bus vs. IEEE 14-bus vs. IEEE 30-bus, all 16-seed means, all four feature sets. Gaps are near zero on both tasks at 5-bus and positive on both tasks at IEEE 14-bus and IEEE 30-bus; Table IV reports uncertainty and multiple-comparisons-aware significance.

summary statistics of the residual/innovation arrays, plus, for localization, a trained per-node classifier ranking every bus’s probability of being the attack target. This feature design is implemented separately from Section III’s but applies the same safeguards. The residual feature is the deployable measurement residual $z - h(\hat{x})$, never the simulator’s hidden ground-truth state; model selection is validation-based rather than test-set-based; non-relational baselines exclude relational columns; and topology_fusion excludes static structural metadata such as node degree, which could otherwise act as a target-eligibility prior because attacks are restricted to load buses. All numbers below use $N=500$ replications and 16 independent seeds at both networks.

C. Cross-Network Result

At 5-bus (Table IV), both 16-seed gaps remain centered near zero with tight confidence intervals (Section IV). At IEEE 14-bus, the detection gap is positive in 11 of 15 non-zero seeds (mean $+0.009 \pm 0.012$) and the localization gap is non-negative in all 16 seeds, zero seeds favoring residual_plus_prior (mean $+0.018 \pm 0.017$). At IEEE 30-bus, the detection gap is positive in all 16 seeds (mean $+0.028 \pm 0.019$) and the localization gap is positive in 12 of 15 non-zero seeds (mean $+0.013 \pm 0.016$). Both tasks show positive mean gaps at both standard IEEE networks.

D. Statistical Evidence at 16 Seeds

At 16 seeds (Table IV), four of the six gaps have a t -based 95% confidence interval excluding zero, with a two-sided t -test rejecting the null at $p < 0.05$ for the same four (IEEE-14

TABLE IV

TOPOLOGY_FUSION VS. RESIDUAL_PLUS_PRIOR (IEEE-14/30: THE STRONGEST NON-RELATIONAL BASELINE IN EVERY SEED), 16 INDEPENDENT SEEDS (500 REPL. EACH), ALL THREE NETWORKS

Network	Task	Mean gap	95% CI	Sign p	t -test p
5-bus	Det.	-0.000 ± 0.009	$[-0.005, +0.004]$	0.500	0.849
5-bus	Loc.	$+0.003 \pm 0.017$	$[-0.006, +0.012]$	0.212	0.500
IEEE-14	Det.	$+0.009 \pm 0.012$	$[+0.002, +0.016]$	0.059	0.011
IEEE-14	Loc.	$+0.018 \pm 0.017$	$[+0.008, +0.027]$	<0.001	<0.001
IEEE-30	Det.	$+0.028 \pm 0.019$	$[+0.018, +0.038]$	<0.001	<0.001
IEEE-30	Loc.	$+0.013 \pm 0.016$	$[+0.004, +0.022]$	0.018	0.007

Sign test excludes exact ties. Using the two-sided t -test p -values, a Bonferroni correction for 6 simultaneous tests requires $p < 0.0083$ (family-wise 95%): three of the four positive IEEE gaps clear it; IEEE-14 detection ($p = 0.011$) does not.

detection $p = 0.011$; IEEE-14 localization $p < 0.001$; IEEE-30 detection $p < 0.001$; IEEE-30 localization $p = 0.007$).

A distribution-free sign test, weaker but assumption-free, agrees for three of the four (IEEE-14 localization $p < 0.001$; IEEE-30 detection $p < 0.001$; IEEE-30 localization $p = 0.018$) but not for IEEE-14 detection specifically ($p = 0.059$, 11 of 15 non-zero seeds positive) – evidence that a small effect relative to per-seed noise can clear a parametric test before it clears the sign test’s coarser bar.

Testing six gaps simultaneously raises a multiple-comparisons concern: a Bonferroni correction requiring $p < 0.0083$ for family-wise 95% confidence still clears IEEE-14 localization, IEEE-30 detection, and IEEE-30 localization, but not IEEE-14 detection. Three gaps (IEEE-14 localization, IEEE-30 detection, IEEE-30 localization) remain statistically significant at the Bonferroni-adjusted threshold. IEEE-14 detection is positive on average but is the weakest-supported IEEE effect, while both 5-bus gaps remain centered near zero ($p = 0.849$ and $p = 0.500$ by t -test). The 16-seed sample is pragmatic rather than based on a formal power analysis; additional seeds or a paired/bootstrap difference analysis would provide tighter, less assumption-dependent bounds on effect magnitude.

E. Interpretation

Across the three test systems, the 16-seed estimates differ more by system than by task: both detection and localization show small positive mean gaps at IEEE-14 and IEEE-30, while both 5-bus gaps remain near zero. These results directly address the limited cross-topology evaluation identified in the 2025 scoping review by Oelhaf et al. [20], while retaining matched test-set sizes and the same 16-seed standard across the three evaluated networks.

VIII. DISCUSSION

At the small custom 5-bus microgrid, explicit topology-relational features show no advantage over a matched non-relational ablation on either detection (mean -0.000 ± 0.009 , 16 seeds) or localization (mean $+0.003 \pm 0.017$). At both standard IEEE test systems, the same ablation shows a small positive gap on both tasks: IEEE 14-bus detection ($+0.009 \pm 0.012$)

and localization ($+0.018 \pm 0.017$); IEEE 30-bus detection ($+0.028 \pm 0.019$) and localization ($+0.013 \pm 0.016$). Both standard IEEE systems therefore differ qualitatively from the small custom microgrid on both tasks.

At 5-bus, the larger improvement comes from combining residual and prior-innovation information rather than from adding topology-relational features (Section IV). Section IV’s redundancy analysis gives the 5-bus null result a structural, not just empirical, footing: on a fixed, small topology, a relational aggregate is close to a linear function of information a flexible classifier already has (mean out-of-fold $R^2 = 0.956$), so there is little left for an explicit relational feature to contribute. If this redundancy argument explained the cross-network pattern, it should weaken at the larger, more meshed IEEE networks – but the same diagnostic, applied to IEEE-14’s relational localization features, gives a mean R^2 of 0.966 (median 0.999, 92% of columns above 0.8), if anything *higher* than 5-bus, not lower. The redundancy hypothesis alone therefore does not explain the cross-network pattern: relational information remains highly linearly recoverable from local features at IEEE-14, yet the explicit relational representation still yields a positive 16-seed localization gap. Linear recoverability does not imply that a fixed-depth tree ensemble will reconstruct the same aggregation reliably from a flat input. The diagnostic therefore rules out missing linear information as the explanation but does not identify the mechanism. An equivalent IEEE-30 redundancy diagnostic is not reported because its detection features use a different, non-comparable representation; developing a separate diagnostic for that representation is left for future work.

One possible explanation is feature discoverability: Hist-GradientBoosting receives a flat input and must infer useful neighbor combinations implicitly, whereas the topology-aware representation supplies those aggregations directly. This interpretation is consistent with the larger-network results but is not tested causally here.

Because the 5-bus and IEEE experiments also differ in measurement/prior settings and feature representation, these between-network differences are descriptive rather than causal; the within-system topology-vs.-matched-ablation comparisons remain the controlled contrasts.

IX. LIMITATIONS

- 1) **Scale and setup dependence:** Three network topologies are evaluated with 16 independent seeds each (Section VII-D). Both tasks show small positive gaps at the standard IEEE systems (14-bus, 30-bus), whereas both 5-bus gaps remain near zero. None uses a real (non-synthetic) network or load profile. With only three networks tested, one custom and two standard, the observed difference is consistent with, but does not establish, a standard-vs-custom or scale/meshedness effect. Additional non-standard networks would be needed to distinguish these explanations.
- 2) **Multiple comparisons and one borderline result:** four of the six gaps in Table IV have a 95% confidence interval excluding zero and $p < 0.05$ by t -test. A Bonferroni correction for six simultaneous tests (requiring $p < 0.0083$ for family-wise 95% confidence) leaves three of four surviving; IEEE-14 detection ($p = 0.011$ uncorrected) is the weakest-supported of the four and does not clear the sign test’s conventional threshold either ($p = 0.059$). Even where statistically significant, effect sizes remain small (0.009–0.028 balanced-accuracy/top-1 points) relative to the much larger gain from combining residual and prior information over either alone (Section IV). The 16-seed sample was not selected by formal power analysis; additional seeds or a paired/bootstrap difference analysis would provide tighter, less assumption-dependent bounds on effect magnitude.
- 3) **Cross-network measurement and prior mismatch:** measurement standard deviations are fixed absolute values shared, unscaled, across all three networks despite their different physical scales, and prior load-forecast uncertainty is 2.5% at 5-bus versus 5% at IEEE-14/IEEE-30 (Section III). These settings were not standardized or swept independently, so comparisons involving 5-bus are not controlled topology- or scale-only experiments. IEEE-14 and IEEE-30 share the same noise/prior convention. A follow-up sensitivity check re-ran the 5-bus ablation at the matching 5% prior uncertainty (16 seeds, same protocol as Section VII): both gaps remained near zero (detection $+0.001 \pm 0.010$, localization $+0.001 \pm 0.017$; neither 95% CI excludes zero), essentially unchanged from the 2.5% setting. This rules out the prior-uncertainty mismatch specifically as the source of 5-bus’s null result, though the measurement-noise scaling and feature-representation differences below remain untested.
- 4) **Cross-network feature representation:** the 5-bus experiments use fixed per-bus graph-level features, whereas IEEE-14/30 use a separate size-agnostic design based on global summaries and per-node localization features (Section VII). This change is necessary to evaluate larger networks, but it means between-network differences can reflect representation as well as network structure. The within-network comparison between `topology_fusion` and its matched `residual_plus_prior` ablation is the controlled quantity.
- 5) **No zero-day generalization** (Section VI): the reported in-distribution results assume attack types represented in training. Under the symmetric withheld-family protocol, both `topology_fusion` and `residual_plus_prior` collapse identically for detection (Table II). Localization was not separately re-tested under the zero-day protocol, and the attack taxonomy is limited to two families.
- 6) **Single testbed family, two attack types:** the present AC-consistent attack generator does not independently vary the compromised-measurement subset; introducing explicit attacker-access constraints (e.g. a fixed budget

of compromisable sensors) as a separate, tunable threat-model parameter is left for future work.

- 7) **Latency results** (Section V) characterize empirical timing on one documented Apple-arm64/macOS 15.6.1 Python/pandapower environment, not hard real-time schedulability or a hardware/RTOS guarantee. The smaller residual_plus_prior feature set was not benchmarked separately; because state estimation dominates measured latency, it is unlikely to change the bottleneck conclusion, but this was not measured directly.

X. CONCLUSION

Across a small custom 5-bus microgrid with steady-state inverter-representative injections and two standard IEEE test systems (14-bus and 30-bus), explicit topology-relational features show different effect sizes under the tested settings. At 16 seeds, both detection and localization show small positive gaps at both standard IEEE systems (IEEE-14: $+0.009 \pm 0.012$ detection, $+0.018 \pm 0.017$ localization; IEEE-30: $+0.028 \pm 0.019$ detection, $+0.013 \pm 0.016$ localization), while the small custom 5-bus microgrid shows no such gap on either task. Because measurement/prior settings and feature representation differ between the 5-bus and IEEE experiments, these cross-network differences should not be interpreted as a causal topology- or scale-only effect (Section VII). Separately, pipeline optimization reduces median latency from 42.98 ms to 16.2–17.0 ms, with p99 below the 20 ms reference in both repeated runs but rare maxima above it; the detector also fails to generalize to attack types withheld from training.

CODE AND REPRODUCIBILITY

Source code, experiment scripts, and instructions to regenerate the reported datasets and results are available at <https://github.com/yagizsarapli/cyber-physical-grid-detection>.

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